

DIKSHANT SHARMA

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🐙 GitHub

📄 Google Scholar

🌐 Website

Education

Central University of Rajasthan, Ajmer

Sep 2022 – Aug 2024

M.Sc.-Computer Scienc (spl.AI & ML), CGPA 8.73/10

Rajasthan, India

Modules: Artificial intelligence, Python, Advanced Algorithm, Machine Learning, Probability Distribution, NLP

University of Rajasthan

Jul 2018 - Oct 2021

B.C.A - Bachelor of Computer Application , Percentage: 73.83 %

Jaipur, Rajasthan

St. Xavier Sr. Sec. School, Jaipur

May 2018

Higher Secondary Education, CBSE (Physics, Chemistry, Maths with I.P.)

Jaipur, Rajasthan

Experience

Research Assistant | Vision Language and Learning Group | IIT Jodhpur

May 2025 - Present

- Advisor : **Dr.Anand Mishra**
- **Vision-Language Models for Autonomous Driving:** Working on a **Bosch** research project in collaboration with **Yasaswi Bharadwaj** (Research Scientist, Bosch) and **Dr. Sahana Prabhu** (IIIT Bangalore), focusing on developing efficient and reliable VLM-based approaches for domain-specific autonomous-driving applications.
- **Multimodal scene-text image retrieval:** Working on multimodal, scene-text aware image-text retrieval models for multilingual settings.
- Served as an Emergency Reviewer for **ICVGIP 2025**, contributing to the peer-review process of computer vision and image processing research submissions.
- Engage in weekly reading-group meetings focused on discussing recent research papers in ML/AI.

Research Assistant | Department of Computer Science | CURAJ

Dec 2023 - July 2024

- Advisor : **Dr.Ajay Indian**
- Designed and implemented a lightweight convolutional neural network (CNN) for multi-class brain tumor classification, optimized to reduce model size while maintaining high classification performance.

Data Science Intern | Mr. S. Bhatia | Pysliq

Jan 2024 - Feb 2024

- Performed comprehensive data collection, preprocessing, analysis, and interpretation to derive meaningful insights for the Number Recognition project.

Publications

- **X-STAR: Cross-lingual Scene Text-aware Image Retrieval**
Dikshant Sharma, Prateek, Anand Mishra
Submitted
- **An improved CNN model for characterizing brain tumors using deep learning**
Ajay Indian, Gaurav Meena*, Dikshant Sharma**
Multimedia Tools and Applications(MTAP)
*(*equal contribution)*

Projects

- **Brain Tumor Detection** : Worked on the classification of various brain tumor diseases by developing a robust deep learning model that achieved 97.24% accuracy. The model demonstrated strong capability in detecting early-stage brain tumors, contributing to improved diagnostic decision support. The project involved designing and training convolutional neural networks and applying advanced deep learning techniques to enhance reliability.
- **Handwritten Number Recognition System** : Developed a robust Handwritten Number Recognition system using the MNIST dataset, comprising 60,000 training and 10,000 testing images of handwritten digits. Implemented convolutional neural networks (CNNs) to effectively capture spatial hierarchies and improve classification performance across digit classes.

- **Stock Price Prediction** : Developed a predictive model for Microsoft stock prices using Long Short-Term Memory (LSTM) networks, leveraging historical market data from 1986 to 2022 to capture long-term temporal dependencies. Conducted analysis of multi-decade stock trends to enhance model performance and provide insights into market behavior.
- **Hotel Management System in PHP** : Developed a web-based system using PHP, HTML, and MySQL, featuring separate admin and user sections. The platform enables administrators to manage room details, perform updates, handle bookings, and process payments, ensuring smooth and efficient system operations.

Relevant Coursework

- **Introduction to large language Model by Mitesh M. Khapra.**
- **CS231N: Deep Learning for Computer Vision** from Stanford University
- **DeepLearning - Mitesh Khapra, SKS Iyengar IIT Ropar & IIT Madras (NPTEL)**

Technical Skills

Programming Languages: Python, SQL, PHP, C, C++

Frameworks/Libraries: PyTorch, Numpy, Sklearn, Matplotlib, Pandas

Technologies: Git, GitHub, LaTeX

Others: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Artificial Intelligence, Hadoop

Certificates & Qualifications

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| • National Eligibility Test - Junior Research Fellowship(CS & Applications)) | <i>December 2024</i> |
| • Graduate Aptitude Test in Engineering(Computer Science Engineering) | <i>February 2025</i> |
| • Two-Week Faculty Development Programme on Deep Learning and Image Processing at Central University of Rajasthan | <i>October 2023</i> |

Languages

- **German** : Two years diploma from University of Rajasthan, October 2021
- **French** : Completed coursework at School
- **English** : Fluent